Python Nested Loop Pattern Printing

MNC / Fresher Interview Practice Worksheet

20 pattern-printing exercises to practise nested loops, row and column logic, spaces, reverse iteration, and conditional statements. Questions only — no solutions included. Exact interview questions vary by company.

Student instructions

• Take n as input wherever applicable.

• Use nested loops to construct each pattern; do not print each row manually.

• Make the pattern work for different input sizes.

• Be ready to explain the outer loop, inner loop, and space logic.

Level 1 — Basic patterns

1. Solid square

Input: n = 4

```
* * * *
* * * *
* * * *
* * * *
```

2. Right-angled triangle

Input: n = 5

```
*
* *
* * *
* * * *
* * * * *
```

3. Inverted triangle

Input: n = 5

```
* * * * *
* * * *
* * *
* *
*
```

4. Increasing number triangle

Input: n = 5

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

5. Repeated row numbers

Input: n = 5

```
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
```

Level 2 — Intermediate interview patterns

6. Right-aligned triangle

Input: n = 5

```
    *
   * *
  * * *
 * * * *
* * * * *
```

7. Continuous number triangle

Input: n = 5

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

8. Alternating binary triangle

Input: n = 5

```
1
0 1
1 0 1
0 1 0 1
1 0 1 0 1
```

9. Hollow square

Input: n = 5

```
* * * * *
*       *
*       *
*       *
* * * * *
```

10. Alphabet triangle

Input: n = 5

```
A
A B
A B C
A B C D
A B C D E
```

Level 3 — Placement practice

11. Full pyramid

Input: n = 5

```
    *
   * * *
  * * * * *
 * * * * * * *
* * * * * * * * *
```

12. Inverted full pyramid

Input: n = 5

```
* * * * * * * * *
 * * * * * * * *
  * * * * * *
   * * *
    *
```

13. Diamond pattern

Input: n = 4

```
    *
   * * *
  * * * * *
 * * * * * * *
  * * * * *
   * * *
    *
```

14. Hollow right-angled triangle

Input: n = 5

```
*
* *
*   *
*     *
* * * * *
```

15. Floyd's triangle

Input: n = 5

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

Level 4 — Logic-building challenges

16. Palindrome number pyramid

Input: n = 5

```
    1
   121
  12321
 1234321
123454321
```

17. Butterfly pattern

Input: n = 4

```
*       *
* *   * *
* * * * * *
* * * ** * * *
* * *  * * *
* *    * *
*       *
```

18. Hollow pyramid

Input: n = 5

```
    *
   * *
  *   *
 *     *
* * * * * * * * *
```

19. Number pyramid

Input: n = 5

```
    1
   1 2 1
  1 2 3 2 1
 1 2 3 4 3 2 1
1 2 3 4 5 4 3 2 1
```

20. X pattern

Input: n = 5

```
*       *
  *   *
    *
  *   *
*       *
```

Practice goal: explain your logic clearly and test your code with more than one input size.